MSAI 713 – Statistical Computing

Master's Practical Assignment

Overfitting, Underfitting and Regularization

This assignment operationalizes concepts from Lecture 5 on:

- Overfitting (High Variance)
- Underfitting (High Bias)
- Bias–Variance Trade-off
- Regularization via $L(\beta) + \lambda P(\beta)$

Dataset Context

We use a structured regression dataset similar to:

Kaggle – House Prices Dataset uploaded for you on this assignment in the LMS

Target Variable: SalePrice
Features: Mixed numerical and categorical predictors.

PART A – Baseline Linear Regression

Task A1 – Fit Ordinary Least Squares (OLS)

```
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error

model = LinearRegression()
model.fit(X_train, y_train)

train_mse = mean_squared_error(y_train, model.predict(X_train))
test_mse = mean_squared_error(y_test, model.predict(X_test))

print("Train MSE:", train_mse)
print("Test MSE:", test_mse)
```

Questions to Students

1. Compare Training MSE and Test MSE.
2. Is the gap small or large?
3. Does the model appear underfit, overfit, or reasonably balanced?
4. Based on Lecture 5 definitions

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, classify the model in terms of:

    - Bias level
    - Variance level

5. Explain why linear regression may struggle if the true relationships are nonlinear.

PART B – Inducing Overfitting

Task B1 – Add Polynomial Expansion

```
from sklearn.preprocessing import PolynomialFeatures

poly = PolynomialFeatures(degree=3)
X_train_poly = poly.fit_transform(X_train)
X_test_poly = poly.transform(X_test)

model.fit(X_train_poly, y_train)

train_mse_poly = mean_squared_error(y_train, model.predict(X_train_poly))
test_mse_poly = mean_squared_error(y_test, model.predict(X_test_poly))

print("Train MSE (Poly):", train_mse_poly)
print("Test MSE (Poly):", test_mse_poly)
```

Questions to Students

1. Why did Training MSE drop drastically?
2. Why did Test MSE increase?
3. Which lecture concept explains this phenomenon?
4. Does this indicate:
    - High bias?

- High variance?

5. Explain mathematically how increasing model complexity affects variance.
6. If we increase polynomial degree further, what do you expect to happen? Why?

PART C – Learning Curves

```
from sklearn.model_selection import learning_curve
import numpy as np
import matplotlib.pyplot as plt

train_sizes, train_scores, val_scores = learning_curve(
   model, X_train_poly, y_train, cv=5,
   scoring='neg_mean_squared_error')

train_error = -train_scores.mean(axis=1)
val_error = -val_scores.mean(axis=1)

plt.plot(train_sizes, train_error, label="Training Error")
plt.plot(train_sizes, val_error, label="Validation Error")
plt.legend()
plt.show()
```

Questions

1. Why does validation error not follow the same downward trend?
2. What does the widening gap indicate?
3. At what point would early stopping be justified?
4. Relate this curve to the bias–variance trade-off .
5. Sketch the learning curve for:
   - An underfit model
   - An overfit model

PART D – Ridge Regularization (L2)

```
from sklearn.linear_model import Ridge
```

```
ridge = Ridge(alpha=10)
ridge.fit(X_train_poly, y_train)

train_mse_ridge = mean_squared_error(y_train, ridge.predict(X_train_poly))
test_mse_ridge = mean_squared_error(y_test, ridge.predict(X_test_poly))

print("Train MSE (Ridge):", train_mse_ridge)
print("Test MSE (Ridge):", test_mse_ridge)
```

Questions

1. Why did training error increase?
2. Why did test error decrease?
3. What role does λ (alpha) play in the objective function?
4. Does Ridge reduce bias or variance more?
5. Why must predictors be standardized before applying Ridge?

PART E – Lasso Regularization (L1)

```
from sklearn.linear_model import Lasso
lasso = Lasso(alpha=0.01, max_iter=10000)
lasso.fit(X_train_poly, y_train)
train_mse_lasso = mean_squared_error(y_train, lasso.predict(X_train_poly))
test_mse_lasso = mean_squared_error(y_test, lasso.predict(X_test_poly))

print("Train MSE (Lasso):", train_mse_lasso)
print("Test MSE (Lasso):", test_mse_lasso)
```

Questions

1. How many coefficients are exactly zero?
2. Why does Lasso perform feature selection?

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3. When might Lasso outperform Ridge?
4. What happens if λ becomes extremely large?
5. Compare interpretability of:
    - OLS
    - Ridge
    - Lasso

PART F – Conceptual Synthesis (Master's Critical Thinking)

Students must write a technical report answering:

1. Define overfitting and underfitting formally
2. Demonstrate empirical evidence from this dataset.
3. Explain how regularization modifies:

$$L(\beta) \rightarrow L(\beta) + \lambda P(\beta)$$

4. Discuss bias–variance implications of increasing λ.
5. Recommend a model for:
    - Small dataset
    - High-dimensional dataset
    - Highly correlated predictors